

3 The acceleration of free fall on the Moon is one-sixth of that on Earth. On Earth it takes time t for a stone to fall from rest a distance of 2 m.

What is the time taken for a stone to fall from rest a distance of 2 m on the Moon?

A

$$\frac{t}{6}$$

B

$$\frac{t}{\sqrt{6}}$$

C

$$t\sqrt{6}$$

D

$$6t$$

Answer: C

$$s = ut + \frac{1}{2}at^2$$

$u = 0$ since the stone falls from rest.

$$2 = \frac{1}{2}at^2$$

4